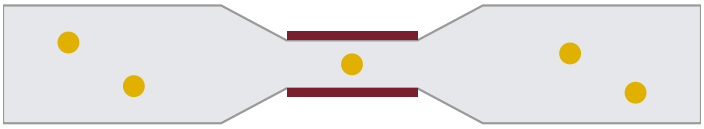
Mini-Lesson 1: Current Lab Pre-Lab

Honors Physics · Circuits · Do with the mini-lesson before your Current Lab class (Tue 11/17 or Wed 11/18). Bring this to lab.

Scan with your iPad camera to open the lesson

Name _____ Period _____

A What is resistance?



resistor = narrow part of the road

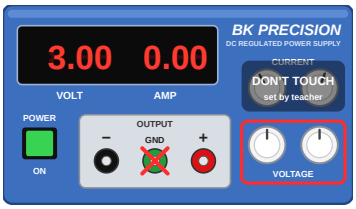
Resistance = how hard it is for _____ to flow. Unit: _____, symbol ____.

Narrower road = _____ resistance = charges flow _____ = _____ current.



The resistance is the number before the ____. This one: _____. "5W" is the _____.

B The power supply



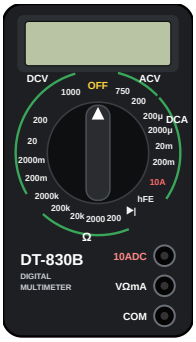
Turn only the bottom _____ knobs.

Never touch the top _____ knobs.

Use the red ____ and black ____ posts, never _____.

Set _____ with nothing plugged in, then build with the power ____.

C Meter as an ammeter (current)



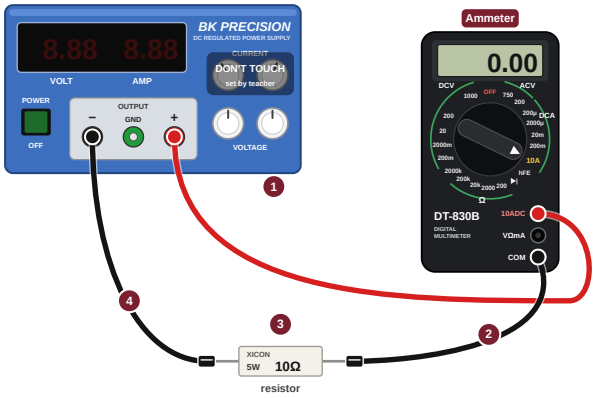
Circle the dial setting you use.

Dial: _____

Red lead: _____ Black lead: _____

Set the dial _____ it's in the circuit. Never turn the dial while _____.

D Worked example: Part 1 circuit



Loop: + post → _____ → _____ → − post. The ammeter is in _____.

On 10A, the meter reading 0.30 means _____. The supply's _____ display should agree.

E Part 2: meter as a voltmeter

	Ammeter (current)	Voltmeter (voltage)
Dial	_____	_____
Red lead	_____	_____
Hook-up	_____	_____

Red lead in 10ADC across a resistor = _____.

Voltages in a loop add: V_{PS} = _____ + _____. 4.00 − 2.50 = _____

F Check questions

1. Dial on 10A. Red lead goes in: _____

2. The ammeter connects: _____

3. V_{PS} = 4.50 V, V_R = 3.00 V. Unknown resistor uses?

work:

V = _____

Big idea in one sentence:

_____

_____